

House Price Prediction – Project Summary

Introduction

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics. Accurate house price prediction is important for buyers, sellers, and real estate businesses as it helps them make informed decisions regarding property valuation. The dataset used in this project contained information about residential properties, including features such as area, number of bedrooms, bathrooms, stories, parking availability, air conditioning, furnishing status, and other amenities.

Data Exploration and Preprocessing

The dataset consisted of 545 records and 13 columns. Initially, the dataset was explored to understand its structure, identify feature types, and check data quality. Missing value analysis revealed that there were no missing values in the dataset. Duplicate record analysis also confirmed that there were no duplicate entries. Several categorical variables such as main road access, guest room availability, basement availability, air conditioning, preferred area, and furnishing status were converted into numerical format using one-hot encoding to make them suitable for machine learning algorithms.

Model Development

The dataset was divided into training and testing sets using an 80:20 ratio. Two regression models were developed and compared:

1. Linear Regression
2. Random Forest Regressor

The performance of both models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics helped measure the accuracy and predictive capability of each model.

Results and Findings

The Linear Regression model achieved an R^2 Score of 0.653, while the Random Forest Regressor achieved an R^2 Score of 0.612. Based on these results, Linear Regression performed slightly better on this dataset and was selected as the preferred model.

Feature importance analysis showed that Area was the most influential factor affecting house prices. Other significant factors included the number of Bathrooms, availability of Air Conditioning, Parking facilities, number of Stories, and Bedrooms. The analysis indicated that larger houses with better facilities and amenities generally command higher market prices.

Visualizations and Insights

Several visualizations were created to better understand the dataset and model performance. A histogram was used to examine the distribution of house prices, a correlation heatmap was generated to identify relationships among variables, and an actual versus predicted price scatter plot was used to evaluate prediction accuracy. These visualizations provided valuable insights into the factors influencing house prices and the effectiveness of the prediction models.

Conclusion

This project successfully demonstrated the application of machine learning techniques for house price prediction. The Linear Regression model provided the best performance among the tested models and explained approximately 65% of the variation in house prices. The findings suggest that property size, number of bathrooms, parking availability, and modern amenities play a major role in determining house value. Real estate businesses can use these insights to improve pricing strategies and better understand customer preferences in the housing market.